SIEGEL'S THEOREM, WITH SOLUTIONS

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This worksheet works through Siegel's two proofs, from [Sil09, §§IX.3–IX.4], that an elliptic curve over a number field K has only finitely many K-rational points with coordinates in $\mathcal{O}_S$, although it can have infinitely many rational points.

Standing notation.

- A number field K with algebraic closure $\overline{K}$, set of places $\mathcal{P}_K$ and set of archimedean places $\mathcal{P}_K^\infty$.
- For $v \in \mathcal{P}_K$, the completion K_v, the absolute value $|\ |_v$ on K_v extending the usual absolute value on $\mathbb{Q}_p$ or $\mathbb{R}$, and the local degree $n_v \coloneqq [K_v : \mathbb{Q}_v]$, where $\mathbb{Q}_v$ is the completion of $\mathbb{Q}$ at the restriction of v. If w is a place of a finite extension of K above v, then $|a|_w = |a|_v$ for $a \in K$.
- The height $H_K(a) \coloneqq \prod_{v\in\mathcal{P}_K} \max(1, |a|_v)^{n_v}$ of $a \in K$, which satisfies $H_K(a) \geqslant 1$ [Sil09, §VIII.5].
- A finite set $S \subseteq \mathcal{P}_K$ containing $\mathcal{P}_K^\infty$. The ring of S-integers $\mathcal{O}_S \coloneqq \{a \in K : |a|_v \leqslant 1 \text{ for all } v \notin S\}$ and its group of units $\mathcal{O}_S^\times$, the group of S-units.

How to use this. Allow about 20 minutes for Problem 1.4 and 40 for Problem 2.3. The two problems are independent. In a first pass, Problem 2.3(6) can wait. Its `Magma` blocks run in seconds in the free online calculator. Everything else is by hand.

1. Integral points and Siegel's theorem

Problem 1.4 is the proof of [Sil09, Corollary IX.3.2.1].

Definition 1.1 (The v-adic distance [Sil09, §IX.2])**.** Let C be a smooth projective curve over K, let $v \in \mathcal{P}_K$, and let $Q \in C(K_v)$. Choose a function $t_Q \in K_v(C)$ that has a zero of order $e \geqslant 1$ at Q and no other zeros. The *v-adic distance* from $P \in C(K_v)$ to Q is

$$d_v(P, Q) \coloneqq \min\big(|t_Q(P)|_v^{1/e}, 1\big),$$

with $d_v(P, Q) \coloneqq 1$ if P is a pole of t_Q.

The distance $d_v(P, Q)$ is small when $P, Q \in C(K_v)$ are v-adically close.

Definition 1.2 (Heights on an elliptic curve [Sil09, §VIII.6])**.** Let E be an elliptic curve over K, and let $f \in K(E)$ be a nonconstant function. The *height relative to f* of a rational point $P \in E(K)$ is $h_f(P) \coloneqq \frac{1}{[K:\mathbb{Q}]} \log H_K(f(P))$, with $h_f(P) \coloneqq 0$ if P is a pole of f. The function f is *even* if $f(-P) = f(P)$ for every $P \in E(\overline{K})$.

With the height h_f of Definition 1.2, Siegel's theorem says that, when $E(K)$ is infinite, the rational points P of an elliptic curve E cannot approach Q quickly, compared with their height: for every $\varepsilon > 0$, all but finitely many of them satisfy $d_v(P, Q) \geqslant \exp(-\varepsilon h_f(P))$.

Theorem 1.3 (Siegel [Sil09, Theorem IX.3.1])**.** *Let E be an elliptic curve over K such that $E(K)$ is infinite. Let $Q \in E(\overline{K})$, let $f \in K(E)$ be a nonconstant even function, and let v be a place of the field $K(Q)$, with $d_v(\cdot, Q)$ defined on E as a curve over $K(Q)$. Then*

$$\lim_{\substack{P\in E(K)\\ h_f(P)\to\infty}} \frac{\log d_v(P, Q)}{h_f(P)} = 0.$$

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Problem 1.4 (Integral points on elliptic curves)**.**
Let E be an elliptic curve over K, given by a Weierstrass equation with coordinate functions x and y.

(1) Show that $t_O := 1/x$ has a zero of order 2 at O and no other zeros. Deduce that, for this choice of t_O, every $v \in \mathcal{P}_K$ and every $P \in E(K_v) \setminus \{O\}$,

$$d_v(P, O) = \max\bigl(1, |x(P)|_v\bigr)^{-1/2}.$$

Solution. **The function x has a pole of order 2 at O and no other poles.** The function x is a coordinate on the affine curve, so it is regular away from O. The map $x\colon E \to \mathbb{P}^1$ has degree 2. Indeed $K(E) = K(x, y)$, and y satisfies a quadratic equation over $K(x)$, so the degree is at most 2. It is not 1, since E has genus one and is not isomorphic to $\mathbb{P}^1$. A nonconstant function has as many poles as its degree, counted with multiplicity [Sil09, Proposition II.2.6]. So x has a pole of order 2 at O, and $1/x$ has a zero of order 2 at O and no other zeros. In Definition 1.1 we may take $t_O = 1/x$ and $e = 2$. If $x(P) \neq 0$, then

$$d_v(P, O) = \min\bigl(|x(P)|_v^{-1/2}, 1\bigr) = \max\bigl(1, |x(P)|_v\bigr)^{-1/2}.$$

If $x(P) = 0$, then P is a pole of t_O, and both sides are 1.

(2) Let $P \in E(K) \setminus \{O\}$ be a rational point with $x(P) \in \mathcal{O}_S$. Show that

$$h_x(P) = \frac{1}{[K : \mathbb{Q}]} \sum_{v \in S} n_v \log \max\bigl(1, |x(P)|_v\bigr),$$

and that some $v \in S$ satisfies $h_x(P) \leqslant -2\,\#S \log d_v(P, O)$, with d_v as in (1).

Solution. **The largest of the $\#S$ terms of the sum is at least their average.** For $v \notin S$, $|x(P)|_v \leqslant 1$, so the factor of $H_K(x(P))$ at v is 1. Taking logarithms gives the formula. Let $v \in S$ be a place where the term $n_v \log \max(1, |x(P)|_v)$ is largest. The sum has $\#S$ terms, so it is at most $\#S$ times this term. As $n_v \leqslant [K : \mathbb{Q}]$, and by (1),

$$h_x(P) \leqslant \#S \log \max\bigl(1, |x(P)|_v\bigr) = -2\,\#S \log d_v(P, O).$$

(3) Show that the set $\{P \in E(K) \setminus \{O\} : x(P) \in \mathcal{O}_S\}$ is finite.

Solution. **Infinitely many such P would be too close to O at one place of S.** If $E(K)$ is finite, there is nothing to prove. So let $E(K)$ be infinite, and suppose that the set is infinite. For each P in the set, (2) gives a place of S. As S is finite, one place $v \in S$ is given for infinitely many P. Let $P_1, P_2, \dots$ be distinct rational points among them, so that

$$h_x(P_i) \leqslant -2\,\#S \log d_v(P_i, O) \quad \text{for all } i.$$

Only finitely many rational points have h_x below a given bound [Sil09, Proposition VIII.6.1]. So $h_x(P_i) \to \infty$, and for large i,

$$\frac{\log d_v(P_i, O)}{h_x(P_i)} \leqslant -\frac{1}{2\,\#S}.$$

The function x is nonconstant and even, and $K(O) = K$. So Theorem 1.3, with $Q = O$ and $f = x$, says that the left side tends to 0, a contradiction.

Remark. In Silverman's proof, "$d_v(P_i, O) \to \infty$" should read "$d_v(P_i, O) \to 0$". The proof gives no bound for the heights of the rational points of the set: Siegel's theorem is proved with Roth's theorem, which gives no bound for its exceptions [Sil09, Remark IX.3.4].

2. The equation $y^2 = f(x)$ and the S-unit equation

Problem 2.3 is the proof of [Sil09, Theorem IX.4.3]. It relies on the following theorem.

Theorem 2.1 (The S-unit equation [Sil09, Theorem IX.4.1])**.** *Let $\lambda, \mu \in K^\times$. Then the equation $\lambda u + \mu w = 1$ has only finitely many solutions $(u, w) \in \mathcal{O}_S^\times \times \mathcal{O}_S^\times$.*

Definition 2.2 collects Silverman's reductions, together with one he omits: the roots of the polynomial are S-integers.

Definition 2.2 (Adapted sets of places)**.** Let $f \in K[T]$ be a polynomial of degree $d \geqslant 1$. The set S is adapted to f if the following conditions hold.

(1) The polynomial f splits over K, as $f = c\prod_{i=1}^{d}(T - \alpha_i)$, with $c \in \mathcal{O}_S^\times$ and $\alpha_1, \dots, \alpha_d \in \mathcal{O}_S$.
(2) For all $i \neq j$, $\alpha_i - \alpha_j \in \mathcal{O}_S^\times$.
(3) The ring $\mathcal{O}_S$ is a principal ideal domain.

Problem 2.3 (The equation $y^2 = f(x)$)**.**
Let $f \in K[T]$ be a polynomial of degree $d \geqslant 3$ with distinct roots in $\overline{K}$. An *S-integral solution* is a pair $(x, y) \in \mathcal{O}_S \times \mathcal{O}_S$ with $y^2 = f(x)$. The goal is to show that there are only finitely many S-integral solutions.

(1) This item reduces the problem to the case where S is adapted to f. Let K' be a finite extension of K, and S' a finite set of places of K' containing the places above S.

(a) Show that S' contains the archimedean places of K', and that $\mathcal{O}_S \subseteq \mathcal{O}_{S'}$.

Solution. **A place of K' outside S' lies above a place of K outside S.** A place w of K' lies above a place v of K, and w is archimedean if and only if v is. If w is archimedean, then $v \in S$, so $w \in S'$. If $a \in \mathcal{O}_S$ and $w \notin S'$ lies above v, then $v \notin S$, so $|a|_w = |a|_v \leqslant 1$. Hence $a \in \mathcal{O}_{S'}$.

(b) Show that K' and S' can be chosen so that S' is adapted to $f \in K'[T]$.

Solution. **Split f, then add finitely many places to S'.** Let K' be a finite extension of K over which f splits, as $f = c\prod_i(T - \alpha_i)$, and start with S' the set of places above S. A nonzero element of K' has absolute value 1 at all but finitely many places of K'. The elements c and $\alpha_i - \alpha_j$, for $i \neq j$, are nonzero, since the roots are distinct. So after adding finitely many places to S', the elements c and $\alpha_i - \alpha_j$ are S'-units and the α_i are S'-integers. These properties persist when S' grows. As the class group of K' is finite, we may enlarge S' once more so that $\mathcal{O}_{S'}$ is a principal ideal domain [Sil09, proof of Proposition VIII.1.6].

Remark. Silverman's reductions omit $\alpha_i \in \mathcal{O}_S$. It does not follow from the others. Over $\mathbb{Q}$ with $S = \{\infty, 2\}$, the roots 1/3, 4/3, 7/3 have differences ± 1 and ± 2, and $\mathbb{Z}[1/2]$ is a principal ideal domain.

(c) Deduce that it is enough to prove that there are only finitely many S-integral solutions when S is adapted to f.

Solution. **Every S-integral solution is an S'-integral solution.** Take K' and S' as in (1)(b), and apply (1)(a).

(2) From now on, S is adapted to f, with c and α_i as in Definition 2.2. Let (x, y) be an S-integral solution. Show that for each i there are $b_i \in \mathcal{O}_S^\times$ and $z_i \in \mathcal{O}_S$ with $x - \alpha_i = b_i z_i^2$.

Solution. **Every prime of $\mathcal{O}_S$ divides $x - \alpha_i$ to an even power.** If $x = \alpha_i$, take $b_i := 1$ and $z_i := 0$. So fix i with $x \neq \alpha_i$. As x and the roots lie in $\mathcal{O}_S$, so do the $x - \alpha_j$. The ring $\mathcal{O}_S$ is a principal ideal domain, by Definition 2.2(3), hence a unique factorization domain. Let π be

a prime element of $\mathcal{O}_S$ that divides $x - \alpha_i$. If π also divided $x - \alpha_j$ for some $j \neq i$, it would divide $\alpha_i - \alpha_j$, which is a unit by Definition 2.2(2). So π divides no other factor of

$$y^2 = c \prod_j (x - \alpha_j),$$

and it does not divide the unit c. In particular $y \neq 0$. Hence the exponent of π in $x - \alpha_i$ is its exponent in y^2, which is even. So every prime occurs to an even power in the factorization of $x - \alpha_i$, and $x - \alpha_i = b_i z_i^2$ with $b_i \in \mathcal{O}_S^\times$ and $z_i \in \mathcal{O}_S$.

(3) Let $L \subseteq \overline{K}$ be the field obtained from K by adjoining a square root of every element of $\mathcal{O}_S^\times$. Show that L is a finite extension of K. Let T be the set of places of L above S, and $\mathcal{O}_T$ the ring of T-integers of L. Choose $\beta_i \in L$ with $\beta_i^2 = b_i$. Show that for all $i \neq j$,

$$\beta_i z_i + \beta_j z_j \in \mathcal{O}_T^\times \quad \text{and} \quad \beta_i z_i - \beta_j z_j \in \mathcal{O}_T^\times.$$

Solution. **Their product is $\alpha_j - \alpha_i$, a unit, and both factors are T-integers.** By Dirichlet's S-unit theorem, $\mathcal{O}_S^\times$ is generated by finitely many elements $\eta_1, \dots, \eta_r$ [Sil09, proof of Proposition VIII.1.6]. Every $b \in \mathcal{O}_S^\times$ is a product of powers of the η_k, so it has a square root in $K(\sqrt{\eta_1}, \dots, \sqrt{\eta_r})$. Hence $L = K(\sqrt{\eta_1}, \dots, \sqrt{\eta_r})$, of degree at most 2^r over K.
Let $w \notin T$ be a place of L, above the place v of K. Then $v \notin S$, so w is non-archimedean, and $|\beta_i|_w^2 = |b_i|_v = 1$ and $|z_i|_w = |z_i|_v \leqslant 1$. By the ultrametric inequality, $|\beta_i z_i \pm \beta_j z_j|_w \leqslant 1$. So both elements lie in $\mathcal{O}_T$. Their product is

$$(\beta_i z_i + \beta_j z_j)(\beta_i z_i - \beta_j z_j) = b_i z_i^2 - b_j z_j^2 = (x - \alpha_i) - (x - \alpha_j) = \alpha_j - \alpha_i,$$

and $|\alpha_j - \alpha_i|_w = 1$ by Definition 2.2(2). Two numbers that are at most 1 and have product 1 are both 1. So $|\beta_i z_i \pm \beta_j z_j|_w = 1$ for every $w \notin T$.

(4) Let $\theta \coloneqq \beta_1 z_1 - \beta_3 z_3$, which is nonzero by (3). Verify *Siegel's identities*

$$\frac{\beta_1 z_1 + \beta_2 z_2}{\theta} - \frac{\beta_2 z_2 + \beta_3 z_3}{\theta} = 1 \quad \text{and} \quad \frac{\beta_1 z_1 - \beta_2 z_2}{\theta} + \frac{\beta_2 z_2 - \beta_3 z_3}{\theta} = 1.$$

Deduce that there is a finite subset of L, depending only on K, S and f, that contains θ for every S-integral solution (x, y) and every choice of the b_i, z_i and β_i.

Solution. **The first terms lie in a finite set, with product $(\alpha_2 - \alpha_1)/\theta^2$.** In both identities, the terms $\beta_2 z_2$ cancel, and the numerators add up to $\beta_1 z_1 - \beta_3 z_3 = \theta$. Let

$$U \coloneqq \{u \in \mathcal{O}_T^\times : 1 - u \in \mathcal{O}_T^\times\}.$$

It is finite, by Theorem 2.1 applied to the number field L, the set T and $\lambda = \mu = 1$, and it depends only on K and S. By (3), every term of the two identities lies in $\mathcal{O}_T^\times$. So

$$u_+ \coloneqq \frac{\beta_1 z_1 + \beta_2 z_2}{\theta} \quad \text{and} \quad u_- \coloneqq \frac{\beta_1 z_1 - \beta_2 z_2}{\theta}$$

lie in U. By the computation in (3),

$$u_+ u_- = \frac{b_1 z_1^2 - b_2 z_2^2}{\theta^2} = \frac{\alpha_2 - \alpha_1}{\theta^2}.$$

So θ is a square root of $(\alpha_2 - \alpha_1)/(uu')$ for some $u, u' \in U$. There are finitely many such elements of L.

(5) Show that there are only finitely many S-integral solutions. Where did the proof use that $d \geqslant 3$?

Solution. **The value of θ determines x. Siegel's identities need three roots.** By the computation in (3), $\theta(\beta_1 z_1 + \beta_3 z_3) = \alpha_3 - \alpha_1$. So

$$\beta_1 z_1 = \frac{1}{2}\Big((\beta_1 z_1 - \beta_3 z_3) + (\beta_1 z_1 + \beta_3 z_3)\Big) = \frac{1}{2}\Big(\theta + \frac{\alpha_3 - \alpha_1}{\theta}\Big),$$

and $x = \alpha_1 + b_1 z_1^2 = \alpha_1 + (\beta_1 z_1)^2$. By (4), θ takes finitely many values, so x does too. For each x, at most two y satisfy $y^2 = f(x)$. The proof used three distinct roots α_1, α_2, α_3 in (4) and here.

Remark. For $d = 2$ the statement is false. The equation $y^2 = 2x^2 + 1$ has infinitely many solutions in integers, given by $y + x\sqrt{2} = (3 + 2\sqrt{2})^k$ for $k \geqslant 0$. The proof is not effective, because Theorem 2.1 is proved with Roth's theorem [Sil09, Remark IX.4.2.2]. Every other step is explicit, and effective bounds for the S-unit equation give effective bounds for the solutions [Sil09, §IX.5].

(6) Let $K = \mathbb{Q}$, $S = \{\infty, 2, 3\}$ and $f = T^3 - 36T = T(T-6)(T+6)$, with $\alpha_1 = 0$, $\alpha_2 = 6$ and $\alpha_3 = -6$. Here $\mathcal{O}_S = \mathbb{Z}[1/6]$.

(a) Show that S is adapted to f. Find $\mathcal{O}_S^\times$, representatives of $\mathcal{O}_S^\times/\mathcal{O}_S^{\times 2}$, and the field L.

Solution. **The units are $\pm 2^m 3^n$, and $L = \mathbb{Q}(i, \sqrt{2}, \sqrt{3})$.** Here $c = 1$, the roots 0, 6 and -6 are integers, and their differences are ± 6 and ± 12. The ring $\mathbb{Z}[1/6]$ is a principal ideal domain: an ideal I is generated by $I \cap \mathbb{Z}$, which is a principal ideal of $\mathbb{Z}$. A rational number is a unit of $\mathbb{Z}[1/6]$ if and only if its numerator and denominator are divisible only by 2 and 3. So $\mathcal{O}_S^\times$ is generated by -1, 2 and 3, the differences of the roots are units, and $L = \mathbb{Q}(\sqrt{-1}, \sqrt{2}, \sqrt{3})$, of degree 8 over $\mathbb{Q}$. The classes of $\mathcal{O}_S^\times$ modulo squares are represented by ± 1, ± 2, ± 3 and ± 6.

(b) For each of the S-integral solutions $(-3, 9)$, $(12, 36)$, $(18, 72)$ and $(294, 5040)$, write $x - \alpha_k = b_k z_k^2$ with $b_k \in \{\pm 1, \pm 2, \pm 3, \pm 6\}$ and z_k a positive integer.

Solution. **The triples (b_1, b_2, b_3) are $(-3, -1, 3)$, $(3, 6, 2)$, $(2, 3, 6)$ and $(6, 2, 3)$.**

(x, y)	x	$x - 6$	$x + 6$
$(-3, 9)$	$-3 = -3 \cdot 1^2$	$-9 = -1 \cdot 3^2$	$3 = 3 \cdot 1^2$
$(12, 36)$	$12 = 3 \cdot 2^2$	$6 = 6 \cdot 1^2$	$18 = 2 \cdot 3^2$
$(18, 72)$	$18 = 2 \cdot 3^2$	$12 = 3 \cdot 2^2$	$24 = 6 \cdot 2^2$
$(294, 5040)$	$294 = 6 \cdot 7^2$	$288 = 2 \cdot 12^2$	$300 = 3 \cdot 10^2$

Remark. The product $b_1 b_2 b_3$ is a square, since $y^2 = b_1 b_2 b_3 (z_1 z_2 z_3)^2$. On the elliptic curve $y^2 = x^3 - 36x$, the map sending a rational point to the classes of (b_1, b_2) modulo squares is the homomorphism of the complete 2-descent [Sil09, Proposition X.1.4]. For instance $(12, 36) = (-3, 9) + (0, 0)$. The homomorphism sends $(0, 0)$ to the classes of $(-1, -6)$, and indeed $3 = (-3)(-1)$ and $6 = (-1)(-6)$.

(c) For the S-integral solution $(12, 36)$, take $\beta_1 = \sqrt{3}$, $\beta_2 = \sqrt{6}$ and $\beta_3 = \sqrt{2}$. The code below computes the terms u and w of the first of Siegel's identities, and their minimal polynomials over $\mathbb{Q}$. Explain how the output shows that $u, w \in \mathcal{O}_T^\times$.

```
R<t> := PolynomialRing(Rationals());
L<i, r2, r3> := NumberField([t^2 + 1, t^2 - 2, t^2 - 3] : Abs);
c1 := 2*r3;  c2 := r2*r3;  c3 := 3*r2;       // beta_k*z_k for (12, 36)
u := (c1 + c2)/(c1 - c3);
w := -(c2 + c3)/(c1 - c3);
u + w;                          // 1
MinimalPolynomial(u);           // t^4 + 8*t^3 + 2*t^2 - 8*t + 1
MinimalPolynomial(w);           // t^4 - 12*t^3 + 32*t^2 - 24*t + 4
```

Solution. **Both are monic over $\mathbb{Z}$, and 1 and 4 are units of $\mathbb{Z}[1/6]$.** An element $\gamma \in L$ that is a root of a monic polynomial $t^n + a_1 t^{n-1} + \cdots + a_n$ with coefficients in $\mathbb{Z}[1/6]$ lies in $\mathcal{O}_T$. Indeed, if $|\gamma|_w > 1$ for some $w \notin T$, then $|\gamma^n|_w > |a_k \gamma^{n-k}|_w$ for all $k \geqslant 1$, which contradicts the ultrametric inequality. So $u, w \in \mathcal{O}_T$. The first polynomial gives $u(u^3 + 8u^2 + 2u - 8) = -1$, so $1/u \in \mathcal{O}_T$. The second gives

$$w(w^3 - 12w^2 + 32w - 24) = -4,$$

so $1/w = -(w^3 - 12w^2 + 32w - 24)/4 \in \mathcal{O}_T$, as 4 is a unit of $\mathbb{Z}[1/6]$. So (u, w) is a solution of $u + w = 1$ in $\mathcal{O}_T^\times$. Note that $1/w$ is not an algebraic integer: the place 2 of S is needed.

Remark. By hand, $u = (2\sqrt{3} + \sqrt{6})/(2\sqrt{3} - 3\sqrt{2})$ and $w = -(\sqrt{6} + 3\sqrt{2})/(2\sqrt{3} - 3\sqrt{2})$, with $(2\sqrt{3} + \sqrt{6})(2\sqrt{3} - \sqrt{6}) = 6 = \alpha_2 - \alpha_1$ and $(2\sqrt{3} - 3\sqrt{2})(2\sqrt{3} + 3\sqrt{2}) = -6 = \alpha_3 - \alpha_1$, as in (3).

(d) The intrinsic `SIntegralPoints` returns all the rational points $P \neq O$ of the elliptic curve $y^2 = x^3 - 36x$ with $x(P) \in \mathbb{Z}[1/6]$, one for each pair $\pm P$. The code below lists their x-coordinates. One of them is not an integer. Find its b_k and z_k.

```
E := EllipticCurve([-36, 0]);
[ Q[1] : Q in SIntegralPoints(E, [2, 3]) ];
// [ -6, -3, -2, 0, 6, 25/4, 12, 18, 294 ]
```

Solution. **For $x = 25/4$, $b_1 = b_2 = b_3 = 1$ and $(z_1, z_2, z_3) = (5/2, 1/2, 7/2)$.** Indeed $25/4 = (5/2)^2$, $25/4 - 6 = (1/2)^2$ and $25/4 + 6 = (7/2)^2$.

Remark. The solutions with $x = 25/4$ are $(25/4, \pm 35/8) = \pm 2(-3, 9)$. Their triple is trivial because they are doubles of rational points.

References

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑